UNIT 8 · TOPIC 7

Machine Selection for a Farm-Equipment Workshop and Its Planning

Syllabus topic: Basic machine selection for a farm equipment workshop and its planning

Basic machines

Machine	Use
LATHE	Turning — cylindrical parts
DRILLING machine	Holes
MILLING machine; SHAPER	Flat surfaces, slots
POWER HACKSAW / bandsaw; SHEARING machine	Cutting stock and sheet
PRESS / press-brake; bending rolls	Sheet work
GRINDERS (bench, tool & cutter, surface)	Finishing, tool sharpening
WELDING sets (arc + gas)	Joining
Power hammer / forge	Forging
Bench/fitting equipment; measuring & inspection gauges (Topic 4b)	Fitting, quality checks

Selection factors: type and volume of products, required accuracy, power availability, capital and running cost, versatility, and operator skill.

Workshop planning and layout

1. Arrange a smooth MATERIAL FLOW: raw material → cutting → forming → welding → machining → finishing → assembly → dispatch, with minimum back-tracking.
2. Group machines by process (job/functional layout) or by product line.
3. Provide adequate space, lighting, ventilation and material handling (cranes, trolleys).
4. Place noisy/hot processes (forging, welding) away from precision machining and inspection.
5. Design for SAFETY — guards, fire points at gas/welding stations, PPE, first aid (Unit 5) — and for future expansion.

> Good layout raises productivity, safety and quality while cutting handling and idle time.

Practice MCQs

Q103. A basic machine for producing cylindrical parts in a workshop is the:

(a) Lathe
(b) Shear
(c) Press-brake
(d) Welding set

Q104. The primary aim of good workshop layout is:

(a) Maximum back-tracking
(b) Smooth material flow with minimum handling and idle time
(c) More noise
(d) Higher scrap

Q105. In workshop planning, forging and welding areas should be:

(a) Next to precision machining and inspection

(b) Separated from precision machining/inspection (heat, noise, fumes)

(c) In the office

(d) Removed entirely

Answers

Q103 — (a) Lathe

Q104 — (b) Smooth material flow with minimum handling and idle time

Q105 — (b) Separated from precision machining/inspection (heat, noise, fumes)

Sources: Hajra Choudhury — Elements of Workshop Technology (Vol I: casting, welding); Kalpakjian — Manufacturing Engineering; AWS welding references; literature on HAZ, weld defects, NDT and sand vs die casting (as given in the source notes, July 2026); NPTEL manufacturing lectures.